RaceDay – API Endpoint Plan (Part 1, Section B)

Roles referenced below: **Organiser**, **Participant**, **Any** (any authenticated user), **None** (public, no login required).

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as either an Organiser or a Participant.	None	`{ fullName, email, password, phoneNumber, role, organiserCode }` (`organiserCode` required only when role is Organiser)	201 Created – user summary (no password) 400 Bad Request – validation failed 403 Forbidden – organiser code missing or invalid 409 Conflict – email already registered
POST	/api/auth/login	Authenticates a user and starts a session, storing the user's ID and role.	None	`{ email, password }`	200 OK – `{ userId, fullName, role }` 401 Unauthorized – invalid credentials
POST	/api/auth/logout	Ends the current user's session.	Any	None	200 OK – logout confirmed

2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the profile of the currently logged-in user.	Any	None	200 OK – user profile 401 Unauthorized – no active session
PUT	/api/users/me	Updates the logged-in user's own profile details.	Any	`{ fullName, phoneNumber, profilePictureUrl }`	200 OK – updated profile 400 Bad Request – validation failed

3. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists events. Optional query filters: `type`, `search`, `upcomingOnly` (default true), `mine` (Organiser's own events).	Any	None	200 OK – array of events
GET	/api/events/{id}	Returns full detail for a single event, including its categories.	Any	None	200 OK – event detail 404 Not Found – event does not exist
POST	/api/events	Creates a new event owned by the logged-in Organiser.	Organiser	`{ name, description, eventDate, location, distanceKm, eventType }`	201 Created – new event 400 Bad Request – validation failed
PUT	/api/events/{id}	Updates an event owned by the logged-in Organiser.	Organiser	`{ name, description, eventDate, location, distanceKm, eventType }`	200 OK – updated event 403 Forbidden – not the owning Organiser 404 Not Found
DELETE	/api/events/{id}	Deletes an event owned by the logged-in Organiser.	Organiser	None	204 No Content 403 Forbidden – not the owning Organiser 404 Not Found

4. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Lists all categories defined for a specific event.	Any	None	200 OK – array of categories 404 Not Found – event does not exist
POST	/api/events/{eventId}/categories	Adds a new age or distance category to an event.	Organiser	`{ name, minAge, maxAge, distanceKm }`	201 Created – new category 403 Forbidden – not the owning Organiser
PUT	/api/categories/{id}	Updates an existing category.	Organiser	`{ name, minAge, maxAge, distanceKm }`	200 OK – updated category 403 Forbidden 404 Not Found
DELETE	/api/categories/{id}	Removes a category from an event.	Organiser	None	204 No Content 403 Forbidden 404 Not Found 409 Conflict – participants already enrolled

5. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/enrolments	Enrols the logged-in Participant into an event under a chosen category.	Participant	`{ categoryId }`	201 Created – enrolment record (status Confirmed) 400 Bad Request – event already taken place 404 Not Found – event/category does not exist 409 Conflict – already enrolled
GET	/api/users/me/enrolments	Returns all events the logged-in Participant has enrolled in.	Participant	None	200 OK – array of enrolments
GET	/api/events/{eventId}/enrolments	Returns all Participants enrolled in an event owned by the logged-in Organiser.	Organiser	None	200 OK – array of enrolments 403 Forbidden – not the owning Organiser

6. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/result	Captures the finish time and position for a Participant's enrolment after the event.	Organiser	`{ finishTime, finishPosition }`	201 Created – result record 400 Bad Request – event has not taken place yet 403 Forbidden – not the owning Organiser 404 Not Found – enrolment does not exist 409 Conflict – result already captured or position already taken
GET	/api/users/me/results	Returns the logged-in Participant's full personal race history.	Participant	None	200 OK – array of results with event name, date, category, finish time, and position
GET	/api/events/{eventId}/results	Returns all captured results for an event owned by the logged-in Organiser.	Organiser	None	200 OK – array of results 403 Forbidden – not the owning Organiser

Part 2 refinements (explained deviations from the first plan)

- **Organiser registration code** – registering as an Organiser requires a secret `organiserCode` configured on the server, so a Participant cannot simply choose the Organiser role.
- **Extra failure codes** – `400` when enrolling after an event or capturing a result before it; `409` when deleting a category that has enrolments or reusing a finishing position within an event.
- **Event list filters** – `type`, `search`, `upcomingOnly` and `mine` query parameters support the Part 3 browse and dashboard pages.
- **Enrolment status** – new enrolments are created as `Confirmed`.